EXECUTIVE BUSINESS CASE & PILOT PROPOSAL

Meter Intelligence Platform

From Anomaly Detection to Confident Action: Unlocking Operational Lift and Revenue Protection

An explainable meter triage engine with automated evidence assembly, human-governed resolution, and rapid ROI for utility operations.

66% Triage Reduction
Cuts manual investigation from 30 min to 10 min

Avoided Truck Rolls
Prevents costly non-fault field dispatches

100% Audit Compliance
Transparent, human-supervised governance

Transforming Smart Meter Alert Spikes into Decisive Business Value

Moving from unmanageable alert queues to automated, evidence-backed human review.

The Current Bottleneck

Operations teams face high volumes of raw meter flags with fragmented context. Analysts spend up to 30 minutes per ticket manually pulling history, weather, and billing records across legacy systems.

- 30 min average triage time
- Severe backlog and analyst fatigue
- Wasteful false-alarm truck rolls

The Intelligence Solution

An automated screening and evidence collation platform. Synthesizes 90-day peer baselines and pre-populates actionable case dossiers in seconds—leaving final judgment to associates.

- 3σ peer-slot baseline filter
- Automated evidence assembly
- 100% human-governed decisions

The Expected ROI

Returns ~333 operational hours per 1,000 cases, eliminates avoidable field dispatches ($150–$250/truck roll), accelerates dispute resolution, and secures unbilled revenue.

- 66% triage efficiency gain
- Capacity unlocked: ~333 hrs / 1k
- Rapid 90-day pilot validation

The High Cost of Inaction in Smart Meter Anomaly Handling

Unfiltered meter signals create severe operational drag, unnecessary truck rolls, and billing exposure.

1. Operational Drag & Triage Fatigue

30+ minutes wasted per anomaly review

Analysts juggle 4–5 legacy portals (AMI data, CIS billing, CRM, weather) just to determine if a spike is legitimate or benign.

2. Avoidable Field Dispatches

$150 – $250 per wasted truck roll

Prematurely escalating unvalidated meter anomalies to field teams wastes substantial fleet budget on non-faulty meters.

3. Delayed Dispute & Billing Resolution

Customer friction & unbilled revenue

Undetected anomalies fester until monthly billing, generating high-friction customer complaints and ombudsman escalations.

4. Scalability Bottleneck

Alert volume outpaces headcount

As smart meter rollouts expand, manual investigation cannot scale linearly without ballooning back-office labor expenditure.

03 / STRATEGIC SOLUTION

An Explainable, Closed-Loop Intelligence Platform

Automating raw data synthesis so operations teams make faster, higher-confidence decisions.

DETECT

Transparent Screening

Filters raw 4-hour telemetry using a 90-day same-slot peer baseline (mean + 3σ). Eliminates seasonal and time-of-day bias before alerts enter the queue.

- Like-for-like slot comparisons
- Minimum 30 peer reads required
- Zero ungrounded noise

INVESTIGATE

AI-Assisted Dossier

Automatically pre-assembles peer history, read continuity, meter diagnostic status, and customer billing markers into an explainable case summary.

- Instant contextual assembly
- Deterministic reasoning rules
- Identifies missing data explicitly

RESOLVE

Human-in-the-Loop

Associates review enriched evidence, confirm root cause, and dispatch field actions or billing adjustments with full audit history retained.

- Associate owns final decision
- Complete audit trail retained
- Seamless CRM/ERP handoff

Governed, Secure, and Audit-Ready from Day One

Clear separation between system evidence, automated reasoning, and associate-owned resolution.

1. INGEST
4h meter telemetry

2. SCREEN
3σ peer-slot baseline

3. ENRICH
Billing & meter status

4. PRESENT
Unified case dossier

5. RESOLVE
Associate sign-off

No Black-Box Hallucinations

Rules are strictly data-grounded and version-controlled. If a data source is missing, the system explicitly reports 'needs-data' rather than asserting unverified causes.

Immutable Audit Trail

Raw meter telemetry, system calculations, and associate notes remain strictly partitioned. Every status update, override, and disposition is logged for compliance.

Enterprise Security

Built with enterprise security in mind: Row-Level Security (RLS), server-side protected credentials, and feature-gated write workflows.

Measurable ROI: Labor Optimization, Field Savings & Revenue Protection

Tangible capacity returned to operations and substantial hard cost avoidance.

333 HRS
Labor Returned / 1k Cases
20 mins saved per anomaly review (from 30m to 10m).

~2.0 FTE
Capacity Unlocked
Equivalent to 2 dedicated analysts freed per 10k alerts handled.

15%–25%
Avoided Truck Rolls
Pre-validation prevents unnecessary dispatches ($150–$250/trip).

< 24 HRS
Faster Dispute Cycle
Resolves billing friction before monthly customer invoices generate.

EXECUTIVE VALUE FRAMEWORK: THREE LEVERS OF IMPACT

1. **Direct Labor Savings:** Automated evidence aggregation reduces investigative drag by 66%, allowing existing teams to absorb smart meter expansion without hiring.
2. **Fleet Cost Avoidance:** Filtering false alarms stops unnecessary truck rolls. Eliminating just 100 unwarranted field visits recovers $18,000–$25,000 in direct fleet expenses.
3. **Revenue Protection:** Rapid identification of under-registering meters prevents revenue leakage while averting high-conflict customer billing disputes.

What the Operations Team Sees: The Unified Case Dossier

Replacing 5 disconnected systems with one clear, evidence-rich decision screen.

CASE SUMMARY: MTR-9824 • USAGE SPIKE (+4.2σ)

Statistical Trigger: Observed 18.4 kWh @ 16:00 vs 4.2 kWh baseline (Threshold: 11.8 kWh).

Read Continuity: PASS — 273 consecutive days unbroken data.

Meter Diagnostics: ACTIVE — Zero tamper or low-battery error flags.

Billing Context: Consistent with past tariff bracket; previous invoices paid on time.

Recommended Next Action: Contact account to confirm recent HVAC/equipment additions prior to booking field service.

OPERATIONAL LIFT FOR ANALYSTS

- **Single-Pane-of-Glass:** No more switching between AMI, billing databases, and CRM portals. All context is pre-fetched and packaged.
- **Prescriptive Guidance:** The platform surfaces recommended next checks, enabling newer associates to execute with senior-level accuracy.
- **Full Associate Ownership:** Associate owns the disposition: Validate, Dispatch Field Tech, Adjust Billing, or Mark Inconclusive.
- **Structured Collaboration:** Notes and comments remain attached to case history, ensuring continuous institutional learning.

Targeted KPIs to Validate Lift During 90-Day Pilot

Rigorous measurement against a baseline cohort before full fleet deployment.

TRIAGE SPEED

< 10 MIN

Baseline: 30 mins

Reduction of ~66% in analyst handling time per investigated anomaly.

PRECISION RATE

> 85%

Candidate Actionability

% of flagged candidates verified as true, actionable operational events.

AVOIDED TRUCKS

20%–30%

Dispatch Reduction

Reduction in unnecessary field dispatches achieved via pre-investigation.

FIRST TOUCH

> 75%

Single-Review Close

Cases resolved on first review without reopening or secondary escalation.

A De-Risked, Staged Rollout Strategy

Structured milestones ensuring operational stability, user adoption, and verified economics.

PHASE 1: HARDEN & CONNECT

Weeks 1 – 4

- Connect live AMI telemetry & billing feeds
- Configure enterprise SSO & Row-Level Security
- Establish pre-pilot operational baselines
- Deliver 2-hour associate training workshop

PHASE 2: 90-DAY PILOT

Weeks 5 – 16

- Deploy with a cohort of 5-8 operations analysts
- Run parallel comparison against standard triage
- Capture override rates and truck roll savings
- Weekly sprint check-ins with operations leads

PHASE 3: ENTERPRISE SCALE

Weeks 17+

- Roll out across full utility meter footprint
- Automate bidirectional ERP/CRM work orders
- Integrate weather & grid telemetry feeds
- Transition to continuous SLA tracking

Decision Request: Sponsor the 90-Day Operational Pilot

Clear mandate, defined resource commitments, and zero operational disruption.

THE DECISION ASK

- **Sponsor a 90-Day Pilot Cohort:** Authorize a controlled pilot with 5-8 operations analysts focusing on a designated smart meter district.
- **Approve Data Access Connections:** Grant read-only API access to AMI telemetry, CIS billing markers, and existing ticketing.
- **Designate an Operations Lead:** Appoint a single operational point-of-contact to review bi-weekly progress and validate scorecard metrics.
- **Zero Disruption Guarantee:** The platform runs in parallel with existing ticketing; current workflows remain intact until benchmarks are proven.

WHY APPROVE TODAY?

- **Immediate Capacity Lift:** Relieve front-line alert fatigue and recapture up to 333 hours per 1,000 cases.
- **Direct Fleet Cost Avoidance:** Curtail unwarranted field dispatches ($150-$250/truck roll) and stop revenue leakage.
- **Auditable Governance:** Deliver compliant, human-supervised AI with zero hallucination exposure and complete audit transparency.
- **Low-Risk Validation:** Full pilot validation without replacing legacy systems or risking customer operations.